Polynomial Continued Fractions: a Proved Logarithmic Ladder, a $4/\pi$ Casoratian Identity, and 482 Irrational Constants

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April 2026

Abstract

We study polynomial continued fractions and prove two main theorems unconditionally. The first establishes a Logarithmic Ladder: a one-parameter quadratic family converges to the reciprocal logarithm for every real parameter greater than one. The second derives a Casoratian telescoping identity yielding $4/\pi$ and a rapidly convergent series for $\pi/4$ with error $O(2^{-N}/N^{7/2})$, from the PCF with $\alpha(n) = -n(2n-3)$ and $\beta(n) = 3n+1$. We prove that this $4/\pi$ family cannot arise from any Gauss ${}_2F_1$ continued fraction, its series representation being forced into a ${}_2F_3$ form over $\mathbb{Q}(\sqrt{5})$. A conditional extension, assuming a conjectural ratio formula (Conjecture 3.3), yields a continuous Gamma-function identity for all non-negative integers m. As a computational application, we certify 482 quadratic generalized continued-fraction constants using Arb ball arithmetic at 10 000-bit precision and depth $N = 5000$, proving irrationality for each by Euler's criterion. PSLQ integer-relation searches against 15 standard constants and 588 Gamma-ratio expressions find no match.

2020 Mathematics Subject Classification: 11J70 (primary), 11Y60, 33C20.
Keywords: continued fractions, irrationality, Casoratian, Gauss hypergeometric, Pincherle's theorem, PSLQ.

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1 Introduction

A *polynomial continued fraction* (PCF) with numerator polynomial $\alpha(n)$ and denominator polynomial $\beta(n)$ is

$$v = \beta(0) + \cfrac{\alpha(1)}{\beta(1) + \cfrac{\alpha(2)}{\beta(2) + \cfrac{\alpha(3)}{\beta(3) + \cdots}}}. \tag{1}$$

The convergents $C_n = p_n/q_n$ are computed via the forward recurrence

$$p_n = \beta(n)\, p_{n-1} + \alpha(n)\, p_{n-2}, \qquad q_n = \beta(n)\, q_{n-1} + \alpha(n)\, q_{n-2}, \tag{2}$$

with $p_{-1} = 1$, $p_0 = \beta(0)$, $q_{-1} = 0$, $q_0 = 1$. Classical work of Wall (1948) and Lorentzen and Waadeland (2008) provides the analytic background on continued fractions, convergence criteria, and the Gauss hypergeometric model relevant here.

The irrationality proof of Apéry (1979) for $\zeta(3)$ showed that polynomial continued fractions can carry deep arithmetic content. More recently, the Ramanujan Machine program (Raayoni et al., 2021; Elimelech et al., 2024) has renewed the search for low-degree examples with closed-form values, and a companion paper (Papanokechi, 2026) develops a spectral classification framework for such families. In this paper we prove unconditional closed evaluations for two explicit one-parameter families: one producing a full logarithmic ladder $1/\ln(k/(k-1))$ for every real $k > 1$, and another yielding $4/\pi$ together with a rapidly convergent series identity for $\pi/4$.

The contributions of this paper are as follows.

Proved results.

1. **Logarithmic Ladder** (Theorem 2.1). The PCF with $\alpha(n) = -kn^2$ and $\beta(n) = (k{+}1)n{+}k$ converges to $1/\ln(k/(k{-}1))$ for all real $k > 1$.

2. **$4/\pi$ Casoratian identity** (Theorems 4.2, 6.3, and 6.4). The PCF with $\alpha(n) = -n(2n{-}3)$, $\beta(n) = 3n{+}1$ converges to $4/\pi$, and the Casoratian telescope yields the series (8) with error $O(2^{-N}/N^{7/2})$.

3. **Gauss CF non-membership** (Proposition 7.1). The $4/\pi$ family is not a Gauss ${}_2F_1$ continued fraction; its series is a ${}_2F_3$ over $\mathbb{Q}(\sqrt{5})$.

4. **482 irrationality certificates** (Proposition 12.2). All 482 quadratic generalized continued-fraction constants $V(A, B, C)$ are proved irrational by Euler's criterion, with Arb-certified enclosures at 10 000-bit precision and PSLQ searches against 15 standard constants finding no match.

Conditional results (assuming Conjecture 3.3).

5. **Gamma identity** (Conditional Theorem 5.1, Corollary 3.5, Corollary 5.4). For all non-negative integers m, the Pi Family PCF converges to $2\,\Gamma(m{+}1)/(\sqrt{\pi}\,\Gamma(m{+}\frac{1}{2}))$, extending the base cases $m = 0, 1$ (which are proved unconditionally).

Numerical/experimental.

6. **Cubic Clausen witness** (Section C; numerical/experimental conjecture). The cubic J-fraction eval_cf$(2, 3, 2, 2/3)$ matches a Clausen square ${}_2F_1(u, 1{-}u; \frac{3}{2}; \frac{27}{64})^2$ with residual below 10^{-341} at 350-digit precision; this match is not claimed as a theorem and is recorded as Open Question 2.

The proofs of the Logarithmic Ladder and the Pi Family base cases ($m = 0, 1$) proceed by closed-form induction on the convergent numerators and denominators, followed by identification of the limiting ratio via classical series. The $4/\pi$ evaluation uses a discrete raising operator L that maps the $m = 0$ solution space to $m = 1$ (Lemma 5.2), combined with Poincaré growth analysis. The Casoratian series identity is derived by a telescoping argument internal to the $m = 1$ recurrence. The Gauss non-membership proof reduces to a polynomial system showing no rational ${}_2F_1$ parameters match.

All numerical results are certified using Arb ball arithmetic via `python-flint` 0.8.0 at 10 000-bit precision (approximately 3 010 decimal digits) and forward depth $N = 5000$. The bracketing method of Lemma 10.2 provides rigorous enclosures: the main proved families achieve bracket widths below 10^{-1500}, yielding over 1500 correct digits. The 482-constant catalogue was generated by a parametric sweep over (A, B, C) with $A \in \{1, \ldots, 5\}$, $|B| \leq 5$, $C \in \{1, \ldots, 5\}$, followed by PSLQ integer-relation searches at 300-digit precision.

The proved statements are separated carefully from the conjectural ones. Section 2 establishes the logarithmic ladder for all real $k > 1$. Sections 3 and 4 prove the base cases $m = 0$ and $m = 1$ of the π-family, and Section 6 derives the Casoratian identity for $\pi/4$. Section 7 proves non-membership in the Gauss continued-fraction class, Section 12 proves irrationality for the quadratic catalogue, and Section C records a cubic Clausen-square witness. Conjecture 3.3 proposes the general ratio law in the parameter m, and Section 5 derives the continuous Gamma quotient from that conjecture together with the proved base cases. Sections 8–11 place the proved formulas in context, while the appendices record numerical certification, the search procedure, exact convergents, and reproducibility data.

2 The Logarithmic Ladder

Theorem 2.1 (Logarithmic Ladder). *For all real $k > 1$, the PCF with $\alpha(n) = -kn^2$ and $\beta(n) = (k{+}1)n + k$ converges to $1/\ln(k/(k{-}1))$.*

Proof. **Claim.**

$$p_n = (n{+}1)!\,k^{n+1}, \qquad q_n = (n{+}1)!\sum_{j=0}^{n}\frac{k^{n-j}}{j+1}. \tag{3}$$

Base case. $p_0 = \beta(0) = k = 1!\cdot k^1$ and $q_0 = 1 = 1!\cdot k^0/1$.

Induction for p_n. Assume $p_{n-1} = n!\,k^n$ and $p_{n-2} = (n{-}1)!\,k^{n-1}$.

$$\begin{aligned}\beta(n)\,p_{n-1} + \alpha(n)\,p_{n-2} &= \big((k{+}1)n + k\big)n!\,k^n - kn^2(n{-}1)!\,k^{n-1}\\ &= n!\,k^n\big((k{+}1)n + k - n\big) = n!\,k^n\cdot k(n{+}1) = (n{+}1)!\,k^{n+1}.\end{aligned}$$

Induction for q_n. Write $S_n = \sum_{j=0}^{n} k^{n-j}/(j{+}1)$ and note $S_n = k\,S_{n-1} + 1/(n{+}1)$. Assume $q_{n-1} = n!\,S_{n-1}$ and $q_{n-2} = (n{-}1)!\,S_{n-2}$. Using $S_{n-1} = k\,S_{n-2} + 1/n$:

$$\begin{aligned}\beta(n)\,q_{n-1} + \alpha(n)\,q_{n-2} &= \big((k{+}1)n{+}k\big)\,n!\big(k\,S_{n-2} + \tfrac{1}{n}\big) - kn^2\,(n{-}1)!\,S_{n-2}\\ &= S_{n-2}\cdot k^2(n{+}1)\,n! + (n{-}1)!\big(kn + k + n\big).\end{aligned}$$

Meanwhile, $(n{+}1)!\,S_n = k^2(n{+}1)\,n!\,S_{n-2} + k(n{+}1)(n{-}1)! + n!$. Since $(n{-}1)!(kn{+}k{+}n) = k(n{+}1)(n{-}1)! + n!$, the two expressions are equal.

Limit. The closed forms (3) are polynomial identities in k, valid for all real k. The convergent ratio is $C_n = k/\sum_{j=0}^{n}(1/k)^j/(j{+}1)$. For $|1/k| < 1$, the series $\sum_{j=0}^{\infty} x^j/(j{+}1) = -\ln(1{-}x)/x$ converges at $x = 1/k$, giving $\sum(1/k)^j/(j{+}1) = k\ln(k/(k{-}1))$. Therefore $\lim C_n = k/(k\ln(k/(k{-}1))) = 1/\ln(k/(k{-}1))$. □

Remark 2.2. The theorem holds for all real $k > 1$, not just integers. As an illustration, using Arb ball arithmetic at 120 decimal digits and forward depth $N = 2000$, we checked 15 non-integer values in $[1.5, 100]$ and found at least 90 matching digits with $1/\ln(k/(k{-}1))$. Table 1 lists representative instances. (This remark is purely illustrative; Theorem 2.1 is proved unconditionally above.)

Table 1: Logarithmic Ladder: first instances

k	$\alpha(n)$	$\beta(n)$	Value
2	$-2n^2$	$3n+2$	$1/\ln 2$
3	$-3n^2$	$4n+3$	$1/\ln(3/2)$
4	$-4n^2$	$5n+4$	$1/\ln(4/3)$
5	$-5n^2$	$6n+5$	$1/\ln(5/4)$
φ	$-\varphi\,n^2$	$(\varphi{+}1)n{+}\varphi$	$1/\ln(\varphi/(\varphi{-}1))$
k	$-kn^2$	$(k{+}1)n{+}k$	$1/\ln(k/(k{-}1))$

3 The Pi Family

3.1 Base case: $2/\pi$

Theorem 3.1 (Pi Family, $m = 0$). *The PCF with $\alpha(n) = -n(2n{-}1)$ and $\beta(n) = 3n + 1$ converges to $2/\pi$.*

Proof. **Claim.**

$$p_n = (2n{+}1)!!, \qquad q_n = (2n{+}1)!! \sum_{j=0}^{n} \frac{j!}{(2j{+}1)!!}. \tag{4}$$

Base case. $p_0 = 1 = (1)!!$ and $q_0 = 1 = 1 \cdot (0!/1!!)$.

Induction for p_n. Assume $p_{n-1} = (2n{-}1)!!$ and $p_{n-2} = (2n{-}3)!!$. Using $(2n{-}1)!! = (2n{-}1)(2n{-}3)!!$:

$$\begin{aligned}\beta(n)\, p_{n-1} + \alpha(n)\, p_{n-2} &= (3n{+}1)(2n{-}1)!! - n(2n{-}1)(2n{-}3)!! \\ &= (2n{-}3)!!\big[(3n{+}1)(2n{-}1) - n(2n{-}1)\big] \\ &= (2n{-}3)!!(2n{-}1)(2n{+}1) = (2n{+}1)!!.\end{aligned}$$

Induction for q_n. Write $T_n = \sum_{j=0}^{n} j!/(2j{+}1)!!$. Note $T_{n-1} = T_{n-2} + (n{-}1)!/(2n{-}1)!!$. With inductive hypotheses $q_{n-1} = (2n{-}1)!!\, T_{n-1}$ and $q_{n-2} = (2n{-}3)!!\, T_{n-2}$:

$$\beta(n)\, q_{n-1} + \alpha(n)\, q_{n-2} = (2n{-}1)!!\big[(3n{+}1)\, T_{n-1} - n\, T_{n-2}\big].$$

Substituting $T_{n-1} = T_{n-2} + (n{-}1)!/(2n{-}1)!!$:

$$(3n{+}1)\, T_{n-1} - n\, T_{n-2} = (2n{+}1)\, T_{n-2} + \frac{(3n{+}1)(n{-}1)!}{(2n{-}1)!!}.$$

The identity reduces to $(3n{+}1)(n{-}1)! = (2n{+}1)(n{-}1)! + n!$, which holds since $n! = n(n{-}1)!$.

Limit. From the closed forms (4), we have $C_n = p_n/q_n = 1/T_n$. The identity (Cuyt et al., 2008, §5.2)

$$\frac{\pi}{2} = \sum_{j=0}^{\infty} \frac{j!}{(2j{+}1)!!} = \sum_{j=0}^{\infty} \frac{2^j (j!)^2}{(2j{+}1)!}, \tag{5}$$

derivable from $\arcsin(1) = \pi/2$, gives $\lim T_n = \pi/2$ and $\lim C_n = 2/\pi$. □

3.2 Parity phenomenon

Remark 3.2 (Parity). For even $c = 2\ell$, the PCF with $\alpha(n) = -n(2n{-}c)$ and $\beta(n) = 3n + 1$ evaluates to an exact rational, since $\alpha(\ell) = -\ell(2\ell{-}2\ell) = 0$ and the CF terminates at depth $\ell{-}1$.

The first even specialisations are listed in Table 2.

Table 2: Pi Family: even parameters (exact rationals)

c	Value	Formula
2	1	1
4	3/2	$\binom{2}{1}/2$
6	15/8	$\binom{4}{2}/2^3$
8	35/16	$\binom{6}{3}/2^4$
10	315/128	$\binom{8}{4}/2^7$
12	693/256	$\binom{10}{5}/2^8$

3.3 Odd parameters and the binomial recurrence

Conjecture 3.3 (Ratio Formula). *Define* $\mathrm{val}(m)$ *as the limit of the PCF with* $\alpha(n) = -n(2n-(2m+1))$ *and* $\beta(n) = 3n+1$. *Then* $\mathrm{val}(m+1) = \mathrm{val}(m) \cdot 2(m+1)/(2m+1)$.

Remark 3.4. Using Arb ball arithmetic at 600 decimal digits and depth $N = 5000$, the recurrence matches to at least 500 digits for $m = 0, 1, \ldots, 10$ and for the half-integer values $m = 0.5, 1.5, \ldots, 10.5$. It is consistent with the closed form $\mathrm{val}(m) = 2^{2m+1}/(\pi\binom{2m}{m})$, since $\binom{2m+2}{m+1} = \binom{2m}{m} \cdot 2(2m+1)/(m+1)$ gives ratio $4\binom{2m}{m}/\binom{2m+2}{m+1} = 2(m+1)/(2m+1)$.

Corollary 3.5 (Full Pi Family (conditional on Conjecture 3.3)). *Assuming Conjecture 3.3, for all* $m \geq 0$ *the PCF converges to* $2^{2m+1}/(\pi\binom{2m}{m})$.

Proof. This is immediate from Theorem 3.1 and repeated application of Conjecture 3.3. Starting from $\mathrm{val}(0) = 2/\pi$, one obtains

$$\mathrm{val}(m) = \frac{2}{\pi} \prod_{j=0}^{m-1} \frac{2(j+1)}{2j+1} = \frac{2^{2m+1}}{\pi\binom{2m}{m}},$$

which is the standard central-binomial product identity. □

Table 3 records the first odd specialisations.

Table 3: Pi Family: odd parameter $c = 2m+1$

m	c	$\alpha(n)$	$\mathrm{val}(m) \cdot \pi$	$\mathrm{val}(m)$
0	1	$-n(2n-1)$	2	$2/\pi$
1	3	$-n(2n-3)$	4	$4/\pi$
2	5	$-n(2n-5)$	$16/3$	$16/(3\pi)$
3	7	$-n(2n-7)$	$32/5$	$32/(5\pi)$
4	9	$-n(2n-9)$	$256/35$	$256/(35\pi)$

4 Convergent Closed Forms

Theorem 4.1 ($m = 0$ convergent numerators). *For the Pi Family with* $m = 0$, $p_n = (2n+1)!! = (2n-1)!! \cdot (2n+1)$.

Proof. Base cases: $p_0=1$, $p_1=3$, $p_2=15$. Induction: assume $p_k = (2k+1)!!$ for $k < n$. The recurrence $p_n = (3n+1)p_{n-1} - n(2n-1)p_{n-2}$ with $P(n) = 2n+1$ reduces to the identity $(2n-1)(2n-3)P(n) = (3n+1)(2n-3)P(n-1) - n(2n-1)P(n-2)$ after dividing by $(2n-3)!!$. Expanding both sides yields $0 = 0$. □

Theorem 4.2 ($m = 1$ convergent numerators). *For the Pi Family with* $m = 1$ *(*$\alpha(n) = -n(2n-3)$, $\beta(n) = 3n+1$*),*

$$p_n = (2n-1)!!\,(n^2 + 3n + 1). \tag{6}$$

The polynomial $h_n = n^2+3n+1$ *(OEIS Foundation, 2024) has discriminant* 5 *and roots* $(-3 \pm \sqrt{5})/2$ *related to the golden ratio.*

Proof. After the substitution $p_n = (2n-1)!!\,u_n$ (using $(2n-1)!! = (2n-1)(2n-3)!!$), the recurrence reduces to

$$(2n-1)\,u_n = (3n+1)\,u_{n-1} - n\,u_{n-2}. \tag{7}$$

Substituting $u_n = n^2+3n+1$, $u_{n-1} = n^2+n-1$, $u_{n-2} = n^2-n-1$:

$$\begin{aligned}\mathrm{RHS} &= (3n+1)(n^2+n-1) - n(n^2-n-1)\\ &= 2n^3 + 5n^2 - n - 1 = (2n-1)(n^2+3n+1) = \mathrm{LHS}.\end{aligned}$$

□

Conjecture 4.3 (General m). *For all $m \geq 0$ and fixed n, the quantity $R(n,m) = p_n(m)/(2n-1)!!$ is a polynomial in m of degree $\lceil n/2 \rceil$.*

A symbolic computation confirms this for $m = 0, \ldots, 12$ and $n \leq 20$ (see the script `conj43_check.py` in the repository, Appendix F). For $m \geq 2$, $R(n,m)$ is rational with denominators dividing $(2n-1)!!$.

5 The Continuous Gamma Identity

The results of §§2–4 are special cases of a single continuous identity. Its extension beyond the base cases $m = 0, 1$ is conditional on Conjecture 3.3. The base cases $m = 0$ and $m = 1$ are proved unconditionally in Steps 1–3 below; the extension to general m in Step 4 assumes Conjecture 3.3.

Conditional Theorem 5.1 (Gamma Function Representation; unconditional for $m = 0, 1$; conditional on Conjecture 3.3 for $m \geq 2$). *The formula below is proved unconditionally for $m = 0$ and $m = 1$. Assuming Conjecture 3.3, for all real $m > -\frac{1}{2}$, the PCF with $\alpha(n) = -n(2n-2m-1)$ and $\beta(n) = 3n + 1$ converges to*

$$\mathrm{val}(m) = \frac{2\,\Gamma(m+1)}{\sqrt{\pi}\,\Gamma(m+\frac{1}{2})}.$$

Proof. We prove the identity for $m = 0$ and $m = 1$ by a purely algebraic argument, then extend to all m via the Wallis reduction formula and Conjecture 3.3 (supported numerically at 600 decimal digits and depth $N = 5000$).

Step 1: The raising operator. Define $L(f)_n = (n+2)\, f_n - (n+1)^2\, f_{n-1}$.

Lemma 5.2. *If f satisfies the m=0 recurrence $f_n = (3n+1)f_{n-1} - n(2n-1)f_{n-2}$, then $g = L(f)$ satisfies the m=1 recurrence $g_n = (3n+1)g_{n-1} - n(2n-3)g_{n-2}$.*

Proof of Lemma. Write $g_{n-1} = (n+1)f_{n-1} - n^2 f_{n-2}$ and $g_{n-2} = n\, f_{n-2} - (n-1)^2 f_{n-3}$. Expanding $(3n+1)g_{n-1} - n(2n-3)g_{n-2}$ and eliminating f_{n-3} via the m=0 recurrence at index $n-1$ yields coefficients

$$\begin{aligned}
&\text{coeff. of } f_{n-1} : (3n^2+4n+1) - n(n-1) = 2n^2+5n+1,\\
&\text{coeff. of } f_{n-2} : -n^2(5n-2) + n(n-1)(3n-2) = -n(2n^2+3n-2),
\end{aligned}$$

matching $g_n = (n+2)[(3n+1)f_{n-1} - n(2n-1)f_{n-2}] - (n+1)^2 f_{n-1}$ identically. □

Since L is linear, it maps the solution space $\mathcal{S}_0$ of the m=0 recurrence into $\mathcal{S}_1$. Checking initial values:

$$\begin{aligned}
&L(P^{(0)})_0 = 2 \cdot 1 - 1 \cdot 1 = 1 = P_0^{(1)}, \quad L(P^{(0)})_1 = 3 \cdot 3 - 4 \cdot 1 = 5 = P_1^{(1)},\\
&L(Q^{(0)})_0 = 2 \cdot 1 - 1 \cdot 0 = 2 = 2\,Q_0^{(1)}, L(Q^{(0)})_1 = 3 \cdot 4 - 4 \cdot 1 = 8 = 2\,Q_1^{(1)}.
\end{aligned}$$

By uniqueness: $P_n^{(1)} = L(P^{(0)})_n$ and $2\,Q_n^{(1)} = L(Q^{(0)})_n$.

Step 2: Poincaré growth analysis. The recurrence has Poincaré characteristic roots $\lambda = 2$ and $\lambda = 1$ (from $\lambda^2 - 3\lambda + 2 = 0$). Both $P_n^{(m)}$ and $Q_n^{(m)}$ are dominant, growing like $(2n)!!$, so $P_n^{(m)}/Q_n^{(m)} \to S^{(m)}$ at rate $O(n!/(2n)!!) = O(2^{-n}/\sqrt{n})$.

Step 3: The limit. From Step 1:

$$S_n^{(1)} = 2 \cdot \frac{(n+2)P_n^{(0)} - (n+1)^2 P_{n-1}^{(0)}}{(n+2)Q_n^{(0)} - (n+1)^2 Q_{n-1}^{(0)}}.$$

Dividing by $(n{+}2)Q_n^{(0)}$ and using $(n{+}1)^2/[(n{+}2)(2n{+}1)] \to \frac{1}{2}$, both correction terms converge to $\frac{1}{2}$, giving $S^{(1)} = 2\,S^{(0)}$.

Step 4: General m and the Gamma identity. The base case $S^{(0)} = 2/\pi$ is established by Theorem 3.1. The Wallis reduction formula gives $I_{m+1} = \frac{2m+1}{2m+2}\,I_m$ where $I_m = \int_0^{\pi/2} \sin^{2m}(x)\,dx = \frac{\sqrt{\pi}\,\Gamma(m+\frac{1}{2})}{2\,\Gamma(m+1)}$. Assuming Conjecture 3.3, the ratio formula $\mathrm{val}(m{+}1)/\,\mathrm{val}(m) = 2(m{+}1)/(2m{+}1) = I_m/I_{m+1}$ propagates inductively from $S^{(0)} = 1/I_0$, yielding $\mathrm{val}(m) = 1/I_m = 2\,\Gamma(m{+}1)/(\sqrt{\pi}\,\Gamma(m{+}\frac{1}{2}))$ for all non-negative integers m. Under the same assumption, the identity extends to all real $m > -\frac{1}{2}$ by analytic continuation. □

Remark 5.3. For $m = 0$ and $m = 1$, Steps 1–3 give a complete algebraic proof independent of Conjecture 3.3: $S^{(0)} = 2/\pi$ and $S^{(1)} = 4/\pi$. For $m \geq 2$, the proof is conditional on the ratio conjecture. No polynomial-coefficient shift operator L_m mapping $\mathcal{S}_m$ to $\mathcal{S}_{m+1}$ has been found for general m.

Corollary 5.4 ((conditional on Conjecture 3.3)). *Assuming Conjecture 3.3, the following specializations hold.*

(i) *Integer $m \geq 0$:* $\mathrm{val}(m) = 2^{2m+1}/\big(\pi\binom{2m}{m}\big)$.

(ii) *Half-integer $m = k{+}\frac{1}{2}$, $k \geq 0$:* $\mathrm{val}(m)$ *is an exact rational (Wallis product partial convergent).*

(iii) $\mathrm{val}(m) \sim \sqrt{4m/\pi}$ *as $m \to \infty$ by Stirling's formula.*

Proof. Part (i) is the integer case of Theorem 5.1. For part (ii), substitute $m = k{+}\frac{1}{2}$ into the Gamma quotient and use $\Gamma(k{+}1) = k!$. Part (iii) is the corresponding Stirling asymptotic for the same quotient. □

6 Casoratian Series Identity for $\pi/4$

We give a Casoratian derivation of the series identity for $\pi/4$. The telescoping argument is internal to the $m = 1$ recurrence, while the identification of the continued-fraction limit as $4/\pi$ comes independently from the Euler/Wallis analysis of the previous section.

6.1 Convergence

Proposition 6.1. *The CF* (1) *with $\alpha(n) = -n(2n{-}3)$, $\beta(n) = 3n{+}1$ converges.*

Proof. By the Śleszyński–Pringsheim theorem (Lorentzen and Waadeland, 2008, Theorem 4.35), the CF converges if $|\alpha(n)| \leq \beta(n)\,\beta(n{-}1)$ for $n \geq 2$. We have $\beta(n)\,\beta(n{-}1) - |\alpha(n)| = (3n{+}1)(3n{-}2) - n(2n{-}3) = 7n^2 - 2$, which is strictly positive for $n \geq 1$. □

6.2 Casoratian recursion

Let $h_n = n^2{+}3n{+}1$ (the polynomial solution from Theorem 4.2) and $g_n = q_n/(2n{-}1)!!$, both satisfying the reduced recurrence (7).

Lemma 6.2 (Casoratian recursion). *The Casoratian $W_n = h_n\,g_{n-1} - h_{n-1}\,g_n$ satisfies $W_n = \frac{n}{2n-1}\,W_{n-1}$ for $n \geq 2$.*

Proof. Substituting the recurrence (7) for h_n and g_n:

$$\begin{aligned} W_n &= \frac{(3n{+}1)h_{n-1} - n\,h_{n-2}}{2n{-}1}\,g_{n-1} - h_{n-1}\,\frac{(3n{+}1)g_{n-1} - n\,g_{n-2}}{2n{-}1} \\ &= \frac{n}{2n{-}1}(h_{n-1}\,g_{n-2} - h_{n-2}\,g_{n-1}) = \frac{n}{2n{-}1}\,W_{n-1}. \end{aligned}$$

□

Theorem 6.3 (Casoratian closed form)**.** *For all $n \geq 1$, $W_n = n!/(2n{-}1)!!$.*

Proof. Base: $W_1 = 5 \cdot 1 - 1 \cdot 4 = 1 = 1!/1!!$. Induction: $W_n = \frac{n}{2n-1} \cdot \frac{(n-1)!}{(2n-3)!!} = \frac{n!}{(2n-1)!!}$. □

6.3 Main series identity

Theorem 6.4. *Let S be the CF with $\alpha(n) = -n(2n{-}3)$, $\beta(n) = 3n{+}1$.*

(i) *S converges, with $|S_n - S| = O(2^{-n}/n^{7/2})$.*

(ii) *$p_n = (2n{-}1)!!\,(n^2{+}3n{+}1)$.*

(iii) *$S = 4/\pi$, equivalently*

$$\frac{\pi}{4} = 1 - \sum_{k=1}^{\infty} \frac{k!}{(2k{-}1)!!\,(k^2{+}3k{+}1)(k^2{+}k{-}1)} = 1 - \sum_{k=1}^{\infty} \frac{2^k}{\binom{2k}{k}(k^2{+}3k{+}1)(k^2{+}k{-}1)}. \tag{8}$$

Proof of (iii). Part (ii) is exactly the closed numerator formula (6) from Theorem 4.2. Part (i) combines Proposition 6.1 with the tail estimate proved immediately below. It remains to establish (iii). The ratio g_n/h_n telescopes:

$$\frac{g_n}{h_n} - \frac{g_{n-1}}{h_{n-1}} = \frac{-W_n}{h_n\, h_{n-1}}.$$

Summing from $k = 1$ to n and using $g_0/h_0 = 1$:

$$\frac{g_n}{h_n} = 1 - \sum_{k=1}^{n} \frac{k!}{(2k{-}1)!!\,(k^2{+}3k{+}1)(k^2{+}k{-}1)}.$$

As $n \to \infty$, the convergents $p_n/q_n = h_n/g_n$ tend to the continued-fraction value S by Proposition 6.1. The value $S = 4/\pi$ is established independently by Lemma 5.2 and Step 3 of the proof of Theorem 5.1: the raising operator L maps $P^{(0)}, Q^{(0)}$ to scalar multiples of $P^{(1)}, Q^{(1)}$, so $S^{(1)} = 2\,S^{(0)} = 4/\pi$ by Theorem 3.1, with no appeal to the present section. Therefore $g_n/h_n \to \pi/4$, and passing to the limit in the telescoping identity yields (8). The alternative form uses $k!/(2k{-}1)!! = 2^k/\binom{2k}{k}$ (see Knuth, 1997, §1.2.6). □

Proof of convergence rate (i). By Stirling, $\binom{2k}{k} \sim 4^k/\sqrt{\pi k}$, so the general term satisfies $t_k \sim \sqrt{\pi k}/(2^k k^4) = O(2^{-k}/k^{7/2})$. Since the tail is dominated by its first term, $|S_N - \pi/4| = O(2^{-N}/N^{7/2})$. At $N = 35$ the error drops below 10^{-16}. □

7 Gauss CF Non-Membership

7.1 Factorisation over $\mathbb{Q}(\sqrt{5})$

Set $\varphi = (1{+}\sqrt{5})/2$, $\psi = (1{-}\sqrt{5})/2$. Since $(k{+}\varphi)(k{+}\psi) = k^2{+}k{-}1$ and $(k{+}2{+}\varphi)(k{+}2{+}\psi) = k^2{+}5k{+}5$, the ratio of consecutive series terms in (8) is

$$\frac{t_{k+1}}{t_k} = \frac{(k{+}1)(k{+}\varphi)(k{+}\psi)}{2(k{+}\frac{1}{2})(k{+}2{+}\varphi)(k{+}2{+}\psi)}. \tag{9}$$

7.2 Pochhammer obstruction

A standard ${}_pF_q\left(\begin{matrix}\cdots\\ \cdots\end{matrix}\middle| z\right)$ has term ratio equal to a product of linear factors $(k + a_i)/(k + b_j)$ with $a_i, b_j \in \mathbb{Q}$. Here the irreducible quadratics $k^2{+}k{-}1$ and $k^2{+}5k{+}5$ (both of discriminant 5) force the parameters into $\mathbb{Q}(\sqrt{5}) \setminus \mathbb{Q}$.

7.3 Gauss CF non-membership

Proposition 7.1. *The CF does not arise from any ${}_2F_1$ ratio via the Gauss continued-fraction theorem.*

Proof. The Gauss CF for ${}_2F_1\left(\begin{smallmatrix} A+1, B \\ C+1 \end{smallmatrix}\middle| z\right) / {}_2F_1\left(\begin{smallmatrix} A, B \\ C \end{smallmatrix}\middle| z\right)$ has unit-denominator coefficients t_k determined by $(A, B, C) \in \mathbb{Q}^3$. Solving $t_1 = c_1$, $t_2 = c_2$, $t_3 = c_3$ (where the c_k are the equivalence-transformed coefficients of our CF) yields a degree-6 polynomial system with six solutions; four are complex and two are real with $C = -3.2590\ldots$ (certified in Arb at 50-digit precision). For these, $|t_4 - c_4| = 5.900\ldots$ (certified), so no match exists. The polynomial system was solved using `SymPy` and all coefficient values verified at 50-digit precision in Arb. □

7.4 ${}_2F_3$ identification

Remark 7.2 (${}_2F_3$ identification). The term ratio (9) formally identifies the series as

$$ {}_2F_3\big(\varphi, \psi;\ \tfrac{1}{2},\ 2{+}\varphi,\ 2{+}\psi;\ \tfrac{1}{2}\big) $$

over $\mathbb{Q}(\sqrt{5})$: the upper parameters $\varphi = (1{+}\sqrt{5})/2$ and $\psi = (1{-}\sqrt{5})/2$ match the factored Pochhammer shifts in (9), while the lower parameters $\frac{1}{2}$, $2{+}\varphi$, $2{+}\psi$ correspond to the denominator factors $(k{+}\frac{1}{2})$, $(k{+}2{+}\varphi)$, $(k{+}2{+}\psi)$. Since both the parameters and the function class depend on the extension $\mathbb{Q}(\sqrt{5})/\mathbb{Q}$, the series does not reduce to any standard ${}_pF_q$ with rational parameters. We state this identification as a structural observation rather than a theorem; a full verification that the Pochhammer products agree term-by-term for all k is a routine but unilluminating exercise in shifted-factorial bookkeeping.

8 Euler CF Duality

The $m = 0$ CF decomposes as $S^{(0)} = \beta(0) + \alpha(1)/T = 1 - 1/T$ where T is the inner tail starting at $n = 1$. Since $S^{(0)} = 2/\pi$ (Theorem 3.1):

Proposition 8.1. $T = \pi/(\pi - 2)$, *equivalently* $T/(T{-}1) = \pi/2$.

Proof. $T = 1/(1 - S^{(0)}) = 1/(1 - 2/\pi) = \pi/(\pi - 2)$. The identity $T/(T{-}1) = \pi/2$ then follows from $T - 1 = (2)/(\pi - 2)$. □

Remark 8.2. This connects our PCF to the classical identity (5), which comes from arcsin(1) (DLMF, 2024, §4.24.2). The partial numerators of the $m = 0$ and $m = 1$ families differ by a linear term: $\alpha_0(n) - \alpha_1(n) = -n(2n{-}1) + n(2n{-}3) = -2n$.

9 Comparison with Brouncker's CF

Lord Brouncker's classical CF for $4/\pi$ is

$$ \frac{4}{\pi} = 1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \cdots}}}, $$

with $\alpha^B(n) = (2n{-}1)^2$ and constant $\beta^B(n) = 2$. Both CFs converge to $4/\pi$ but differ structurally:

	Brouncker	**Novel** (m=1)
$\alpha(n)$	$(2n-1)^2$	$-n(2n-3)$
$\beta(n)$	2 (constant)	$3n+1$ (linear)
p_n	$(2n+1)!!$	$(2n-1)!!(n^2+3n+1)$
Series	Leibniz $\sum(-1)^k/(2k+1)$	Eq. (8)
Rate	$O(1/n)$	$O(2^{-n})$

	Brouncker			**Novel**		
n	p_n	q_n	S_n	p_n	q_n	S_n
0	1	1	1.000000	1	1	1.000000
1	3	2	1.500000	5	4	1.250000
2	15	13	1.153846	33	26	1.269231
3	105	76	1.381579	285	224	1.272321
4	945	789	1.197719	3045	2392	1.272993
5	10395	7734	1.344065	38745	30432	1.273166
8	34459425	28018665	1.229874	180405225	141690240	1.273237
						$4/\pi \approx 1.273\,239\,545$

At $n = 8$, Brouncker's error is $\sim 4 \times 10^{-2}$ while the novel CF error is $\sim 3 \times 10^{-6}$, a factor of 10^4. The two CFs are not related by any equivalence transformation, contraction, or Möbius substitution.

10 Convergence Theory

10.1 Worpitzky condition

Lemma 10.1. *For both families, the normalised partial numerators $|\tilde{\alpha}_n| = |\alpha_n|/(\beta_n\,\beta_{n-1})$ satisfy $|\tilde{\alpha}_n| < 1/4$ for all sufficiently large n.*

Proof. **Log family.** $|\tilde{\alpha}_n| \to k/(k+1)^2 < 1/4$ for $k > 1$ (since $(k-1)^2 > 0$).

Pi family. $|\tilde{\alpha}_n| = n(2n-1)/((3n+1)(3n-2)) \to 2/9 < 1/4$. At $n = 1$: $|\tilde{\alpha}_1| = 1/4$ (borderline); strict inequality holds for $n \geq 2$, and convergence at equality at finitely many n follows from the classical extension (Lorentzen and Waadeland, 2008, §4.2). □

10.2 Bracketing

Lemma 10.2. *If $\alpha_n < 0$ and $\beta_n > 0$ for all $n \geq 1$, and the CF converges to L, then* $\min(C_N, C_{N+1}) \leq L \leq \max(C_N, C_{N+1})$.

Proof. The determinant $p_n q_{n-1} - p_{n-1} q_n = \prod_{j=1}^{n}(-\alpha_j) > 0$ shows $C_n - C_{n-1}$ alternates in sign. See Lorentzen and Waadeland (2008, Theorem 4.35). □

11 Meta-Family

Remark 11.1 (Unified template). The template $\alpha(n) = -(\alpha_0 n^2 + \beta_0 n)$, $\beta(n) = \gamma_0 n + \delta_0$ contains both the Logarithmic Ladder ($\beta_0 = 0$, $\delta_0 = \alpha_0$, $\gamma_0 = \alpha_0 + 1$) and the Pi Family ($\alpha_0 = 2$, $\gamma_0 = 3$, $\delta_0 = 1$) as special cases. Two additional sub-families, $(\alpha_0, \beta_0, \gamma_0, \delta_0) = (1, 0, 4, 2) \to 2/\ln 3$ and $(4, 0, 8, 4) \to 4/\ln 3$, were found and certified at 10 000-bit precision with depth $N = 5000$; both are instances of the Logarithmic Ladder at $k = 3$ (and its equivalence-transformed double).

12 Quadratic GCF Constants

Define $V(A,B,C) = 1 + \mathbf{K}_{n\geq 1}\, 1/(An^2+Bn+C)$.

Lemma 12.1 (Denominator growth)**.** *Let $A \geq 1$, $C \geq 1$, and $An^2+Bn+C > 0$ for $n \geq 1$. Then the convergent denominators $Q_n = (An^2+Bn+C)\,Q_{n-1} + Q_{n-2}$ satisfy $\log Q_n \geq n\log n - O(n)$.*

Proof. Since $Q_{n-2} \geq 0$ and $Q_0 = 1$, $Q_1 = A+B+C \geq 1$, we have $Q_n \geq (An^2+Bn+C)\,Q_{n-1}$ for all $n \geq 1$. Iterating:

$$Q_n \geq \prod_{j=1}^{n}(Aj^2+Bj+C).$$

For $j \geq j_0$ with j_0 depending only on A,B,C, we have $Aj^2+Bj+C \geq j^2/2$, so

$$\log Q_n \geq \sum_{j=j_0}^{n} \log(j^2/2) + O(1) = 2\sum_{j=j_0}^{n} \log j - (n-j_0+1)\log 2 + O(1).$$

By Stirling, $\sum_{j=1}^{n} \log j = n\log n - n + O(\log n)$, giving $\log Q_n \geq 2n\log n - O(n)$, and in particular $\log Q_n \geq n\log n$ for all sufficiently large n. □

Proposition 12.2. *For all (A,B,C) with $A \geq 1$, $C \geq 1$, and $An^2+Bn+C > 0$ for $n \geq 1$, $V(A,B,C)$ is irrational.*

Proof. The Wronskian identity $|P_nQ_{n-1} - P_{n-1}Q_n| = 1$ holds for all n (since all partial numerators equal 1). By Lemma 12.1, Q_n grows superexponentially, so $|V - P_n/Q_n| = 1/(Q_nQ_{n+1}) = o(Q_n^{-1})$. Euler's irrationality criterion then gives irrationality of V. □

Our systematic scan found **482 distinct** such constants. All values in this paragraph were first computed at 300 decimal digits with forward depth $N = 2000$, and the final Arb enclosures are recorded in Appendix A. A PSLQ (Ferguson et al., 1999) integer relation search at 300-digit precision over 15 standard constants $\{\pi, \pi^2, e, \ln 2, \ln 3, \gamma, G, \zeta(3), \zeta(5), \sqrt{2}, \sqrt{3}, \sqrt{5}, \varphi, \sqrt{\pi}, \pi\ln 2\}$ finds no relation for any V-constant, and none is algebraic of degree ≤ 8 with coefficients $\leq 10^5$. Pairwise cross-relation tests on 91 representative constants (435 pairs, PSLQ with coefficients up to 500) and triple-relation tests (455 triples, coefficients up to 200) all return negative. A targeted sweep against 588 expressions built from Gamma ratios and Pi Family values also produces no matches.

These computations have a mathematical role beyond data collection. Proposition 12.2 gives a uniform irrationality theorem for the quadratic family, and the negative PSLQ evidence indicates that the 482 values are not simple reparameterisations of the standard constants already in the search basis.

The only structural relation found is the *shift symmetry* $V(1,B,C) - V(1,-B,C) \in \mathbb{Z}$ for all tested B,C, following from the palindromic structure of the GCF denominators. This gives strong (but non-rigorous) evidence of mutual algebraic independence.

Remark 12.3. An earlier PSLQ search returned $[0,1,-2,1]$ relative to $\{V, \sqrt{5}, \varphi, 1\}$; this is the trivial identity $\sqrt{5} - 2\varphi + 1 = 0$ with zero coefficient on V.

A Numerical Certification

All results are certified using Arb ball arithmetic (`python-flint` 0.8.0, 10 000-bit precision) at depth $N = 5000$. By Lemma 10.2, the limit lies in $[\min(C_N, C_{N-1}), \max(C_N, C_{N-1})]$. Table 4 summarises the main certified enclosures. These computations serve as certification of the proved formulas and of the catalogue values recorded in Section 12.

Table 4: Arb-certified intervals (depth 5000)

PCF	Target	Bracket width	Digits
$\alpha{=}{-}2n^2,\ \beta{=}3n{+}2$	$1/\ln 2$	$< 10^{-1509}$	1509
$\alpha{=}{-}3n^2,\ \beta{=}4n{+}3$	$1/\ln(3/2)$	$< 10^{-2265}$	2265
$\alpha{=}{-}n(2n{-}1),\ \beta{=}3n{+}1$	$2/\pi$	$< 10^{-1508}$	1508
$\alpha{=}{-}n(2n{-}3),\ \beta{=}3n{+}1$	$4/\pi$	$< 10^{-1518}$	1518

For the $m = 1$ CF, backward recurrence at depth 500 with 80-digit arithmetic confirms

$$S = 1.273239544735162686151070106980114896275677165923651589 98\ldots$$

matching $4/\pi$ to all 60 digits.

B Discovery Methodology

1. **MITM sweep.** Fix $\beta(n) = dn{+}e$; enumerate $\alpha(n) = pn^2{+}qn{+}r$ with $|p|, |q|, |r| \leq 5$; match PCF values against a database of 200 known constants.
2. **Modular signature sweep.** Vary $d \in \{2, \ldots, 8\}$, $e \in \{1, \ldots, 8\}$. The $d{=}3$ family produced both π- and ln-hits.
3. **Parametric generalisation.** Sweep c in $\alpha(n) = -n(2n{-}c)$ and k in $\alpha(n) = -kn^2$, discovering the full infinite families.

C Experimental Cubic and Quartic Clausen Witnesses

The proved families above motivated a parallel high-precision search in nearby cubic J-fractions. The numerically strongest cubic case is the one-parameter family

$$B(\beta) = \beta + \mathbf{K}_{n\geq 1} \frac{2n^3}{2n+\beta}, \qquad \beta \in \mathbb{Q}_{>0}.$$

At the anchor point $\beta = \frac{2}{3}$,

$$\begin{aligned} B_{\text{cubic}} &= \text{eval_cf}(2, 3, 2, 2/3) \\ &= \frac{2}{3} + \mathbf{K}_{n\geq 1} \frac{2n^3}{2n+\frac{2}{3}} = 1.110030230754775938693\ldots. \end{aligned}$$

With $\theta = \arcsin(3\sqrt{3}/8)$, this matches

$${}_2F_1\big(u, 1-u; \tfrac{3}{2}; \tfrac{27}{64}\big)^2 = \left(\frac{\sin(\lambda\theta)}{\lambda \sin\theta}\right)^2, \qquad \lambda = 1 - 2u,$$

for $u = 0.18852361548552202250\ldots$, with hypergeometric residual below 10^{-341} in independent `mpmath` checks at 350-digit precision.

A companion evaluation at $\beta = \frac{1}{3}$ gives

$$B\big(\tfrac{1}{3}\big) = 0.805664740414447796515\ldots = {}_2F_1\big(u', 1-u'; \tfrac{3}{2}; \tfrac{27}{64}\big)^2,$$

with $u' = -0.24589471253854869965\ldots$ on the $\lambda > 1$ sheet of the same cubic modular point $z = 27/64$.

Remark C.1. Exhaustive integer-relation searches at 300-digit precision, including `findpoly` to degree 20, PSLQ over mixed bases involving $\sqrt{3}$ and the elliptic periods $K(k), K(k')$ with $k = 3\sqrt{3}/8$, and pairwise combination tests over all five computed λ-values, find no algebraic structure for the deformation parameter $\lambda(\beta)$.

Remark C.2 (Structural map of the point 27/64). The point $z = 27/64$ is the ramification point of the cubic Chebyshev resolvent $4x^3 - 3x + \frac{8}{3\sqrt{3}}\sqrt{z} = 0$, whose discriminant $\Delta_x = -16(64z - 27)$ vanishes exactly at $z = 27/64$. In the variable $t = \sqrt{1 - 64z/27}$, the pulled-back Gauss equation for ${}_2F_1(u, 1-u; \frac{3}{2}; z)$ acquires a regular singularity at $t = 0$. The integers $27 = 3^3$ and $64 = 4^3$ arise from the cubic Pochhammer combinatorics $(\frac{1}{3})_n(\frac{2}{3})_n = (3n)!/(27^n n!)$ and the Chebyshev normalization, respectively.

The real branch persists up to a numerically observed endpoint $\beta_{\text{crit}} \approx 0.7478702596$ where $\lambda(\beta_{\text{crit}}) = 0$ and the value saturates at $(\theta/\sin\theta)^2 \approx 1.18466547096$. The associated j-invariant $j(27/64) = 29704593673/15968016$ is rational but not an algebraic integer, so $z = 27/64$ is not a classical CM singular modulus; the point behaves as a *modular shadow* rather than a genuine cusp of the continued fraction.

A preliminary scan at the quartic point $z = \frac{1}{4}$ for the family $\text{eval_cf}(1, 4, 7, \beta)$ shows a weak shadow at five-digit precision; this is recorded as a direction for future investigation in §D.

D Open Questions

The following questions arise directly from the proved results and numerical evidence.

1. **Prove Conjecture 3.3.** The ratio formula $\text{val}(m+1) = \text{val}(m) \cdot 2(m+1)/(2m+1)$ is verified numerically to at least 500 digits for $m = 0, 1, \ldots, 10$ and half-integer values (Section 3). A proof would upgrade all conditional results in this paper (Conditional Theorem 5.1, Corollaries 3.5 and 5.4) to unconditional theorems and complete the continuous Gamma-function identity for all m. This is the primary open problem of the paper.

2. **Proof of the Clausen-square identity.** Prove that $B(2/3) = [{}_2F_1(u, 1-u; \frac{3}{2}; \frac{27}{64})]^2$ for $u = 0.18852361548552202250\ldots$ as an exact identity, not merely a high-precision numerical coincidence. The classical Clausen identity provides a structural scaffold; what is missing is a direct derivation connecting $\text{eval_cf}(2, 3, 2, \beta)$ to the hypergeometric value at this argument.

3. **Nature of the deformation parameter $\lambda(\beta)$.** For each β in the real branch $[0, \beta_{\text{crit}}]$, the map $\beta \mapsto \lambda(\beta)$ is smooth and monotone decreasing. Integer-relation searches at 300-digit precision find no algebraic relation for $\lambda(\beta)$ at degree ≤ 20 with coefficient height $\leq 10^5$, no relation involving $\sqrt{3}$ or the elliptic periods $K(k), K(k')$ for $k = 3\sqrt{3}/8$, and no relation of the form $\lambda(\beta)\theta/\pi \in \mathbb{Q}$. Whether $\lambda(\beta)$ is transcendental for all rational β, or algebraic at a sparse set of special values, remains open.

4. **Non-classical modular status and the quadratic catalogue.** The elliptic modulus $k = 3\sqrt{3}/8$ satisfies $k^3 + (k')^3 = 0.71359\ldots \neq 1$, placing $z = 27/64$ strictly outside the Borwein signature-3 singular locus. The associated nome $q = 0.034194\ldots$ matches no entry $q_r = e^{-\pi\sqrt{r}}$ for $r \leq 36$. What modular or arithmetic object, if any, organizes the selection of $z = 27/64$ by the continued-fraction family $\text{eval_cf}(2, 3, 2, \beta)$? In the quadratic catalogue, the constant $V(3, 1, 1)$ is characterized in a companion paper (Papanokechi, 2026) as a confluent-Heun / Stokes obstruction within the spectral classification of polynomial continued fractions; the precise relationship between these two obstruction mechanisms remains to be clarified.

5. **The real-branch endpoint.** The map $\beta \mapsto V(\beta)$ appears to have a critical point near $\beta_{\text{crit}} \approx 0.74787025960$, where $V(\beta_{\text{crit}}) \approx (\theta/\sin\theta)^2$, the $\lambda \to 0$ limit of the trigonometric form. Does this endpoint admit a closed form, and does the family admit a natural complex or hyperbolic continuation beyond it?

6. **Higher-degree analogues.** Does an analogous Clausen-square collapse occur for $\text{eval_cf}(p, d, \alpha, \beta)$ with $d \geq 4$? A preliminary scan at the quartic point $z = 1/4$ for $\text{eval_cf}(1, 4, 7, \beta)$ shows a weak match at five-digit precision; this deserves systematic high-precision investigation.

7. **Casoratian telescoping for higher-rank recurrences.** Determine whether the Casoratian telescoping method of Section 6 extends to higher-rank recurrences ($d \geq 2$) to yield closed-form evaluations beyond the logarithmic and π sectors.

E Exact Convergents

Log (k=2): $p_n = (n{+}1)! \cdot 2^{n+1}$.

n	p_n	q_n	C_n
0	2	1	2.000000
1	8	5	1.600000
2	48	32	1.500000
3	384	262	1.465649
4	3840	2644	1.452345
5	46080	31832	1.447584

Limit: $1/\ln 2 = 1.442695\ldots$

Pi (m=0): $p_n = (2n{+}1)!!$.

n	p_n	q_n	C_n
0	1	1	1.000000
1	3	4	0.750000
2	15	22	0.681818
3	105	160	0.656250
4	945	1464	0.645492
5	10395	16224	0.640688

Limit: $2/\pi = 0.636620\ldots$

Pi (m=1): $p_n = (2n{-}1)!!\,(n^2{+}3n{+}1)$.

n	p_n	q_n
0	1	1
1	5	4
2	33	26
3	285	224
4	3 045	2 392
5	38 745	30 432
6	571 725	449 040
7	9 594 585	7 535 616
8	180 405 225	141 690 240

Limit: $4/\pi = 1.273239\ldots$

F Reproducibility

Python 3.12, `mpmath` 1.4.1, `python-flint` 0.8.0 (Arb), `sympy` 1.14.0. Forward recurrence (2) with p_{-1}=1, p_0=$\beta(0)$, q_{-1}=0, q_0=1. Runtime: <2 hours on a standard desktop. The symbolic verification of Conjecture 4.3 (for $m = 0, \ldots, 12$ and $n \leq 20$) is performed by `conj43_check.py` in the repository.

Acknowledgements

Inspired by the Ramanujan Machine (Raayoni et al., 2021; Elimelech et al., 2024). Computations performed with `mpmath` and `python-flint` (Arb).

Computational Tools. This research was conducted using the SIARC (Self-Iterating Analytic Relay Chain) v6.5 methodology, employing AI language models as research assistants throughout the discovery and verification workflow. Specific tools used include Anthropic Claude (claude.ai) and GitHub Copilot (GPT-4 backend) for hypothesis generation, symbolic computation review, proof scaffolding, and manuscript preparation. All mathematical claims, numerical values, and proof steps were independently verified by the author using certified Arb ball arithmetic (`python-flint` 0.8.0), `mpmath` 1.4.1, and `SymPy` 1.14.0, with no mathematical content sourced from AI output without independent verification.

The epistemic governance framework governing AI use in this research is documented in: (i) the AEAL (Agent Epistemic Accountability Layer) framework (Papanokechi, 2026a); (ii) the ZTEK (Zero Trust Epistemic Kernel) protocol (Papanokechi, 2026b); and (iii) the AI Behavior Science founding paper (Papanokechi, 2026c). This disclosure follows *Experimental Mathematics* editorial policy on AI-assisted research.

Data Availability

All computation scripts and Arb-certified output files are available at `https://github.com/papanokechi/pcf-research`.

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